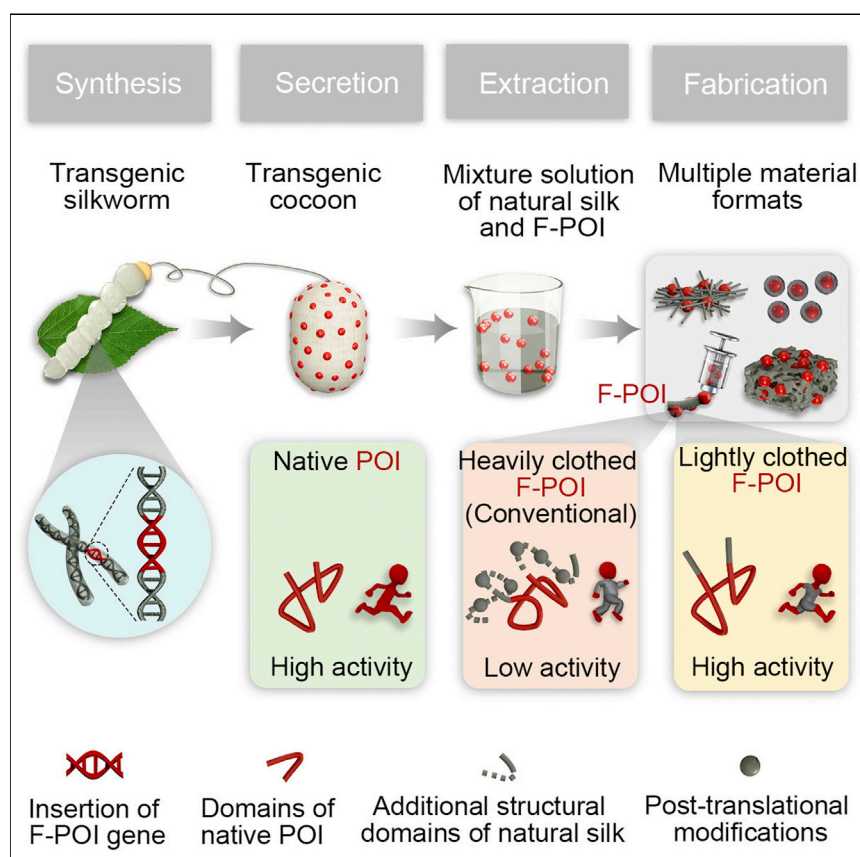


Article

Genetic hybridization of highly active exogenous functional proteins into silk-based materials using “light-clothing” strategy



To ensure successful genetic hybridization (synthesis, secretion, extraction, and fabrication) of exogenous protein(s) of interest (POI) into various silk-based materials using transgenic silkworms, additional structural domains from natural silk are routinely fused with POI to form fusion POI (F-POI). These additional domains lead to superfluous post-translational modifications, compromise normal folding, and severely impair the activity of F-POI. Here, a “light-clothing” strategy was proposed to remarkably enhance F-POI activity by rationally trimming redundant parts of additive domains without compromising the hybridization process.

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Highlights

Exogenous proteins are genetically hybridized into silk materials

Trimming redundant parts of additive domains enhances exogenous protein activity

This trim does not compromise the genetic hybridization process



Demonstrate

Proof-of-concept of performance with intended application/response

Long et al., Matter 4, 2039–2058
June 2, 2021 © 2021 Elsevier Inc.
<https://doi.org/10.1016/j.matt.2021.03.020>



Article

Genetic hybridization of highly active exogenous functional proteins into silk-based materials using “light-clothing” strategy

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SUMMARY

Genetic hybridization of exogenous functional protein(s) of interest (POI) with silk using transgenic silkworms provides promising bio-synthesis platforms to design silk-based materials. To ensure successful hybridization, additive domains from natural silk are routinely fused with POI to form fusion POI (F-POI), but they often lead to superfluous post-translational modifications, compromise normal folding, and severely impair the activity of POI. Here, “light-clothing” strategy, i.e., rationally trimming redundant parts of additive domains, is proposed. As visualized by green fluorescent protein as marker POI, this strategy avoids supernumerary post-translational modifications in F-POI synthesis, and enables its normal secretion and extraction. Applying epidermal growth factor as model functional POI, material fabrication versatility of the extracted mixture of F-POI and silk is demonstrated. This strategy increases structural similarity between F-POI and native POI and remarkably enhances F-POI activity to the same level as commercial POI, which facilitates applications of these advanced bio-synthesis platforms for developing high-performance materials.

INTRODUCTION

During the past decades, silks, produced by sericulture or bioengineered silk via fermentation, have expanded their applicability from the millennia-old textile industry to various emerging areas, such as electronics, photonics, energy, environmental science, and biomedicine.¹ Genetic hybridization of exogenous protein(s) of interest (POI) into silk-based materials using transgenic silkworm (*Bombyx mori*) as a higher eukaryotic bio-hybridization platform provides a promising bottom-up approach to design, synthesize, and functionalize these materials.^{2–5} The transgenic silkworm techniques hold exciting opportunities to combine recent progress in modern bioengineering with advantages of traditional farming, i.e., production of proteins at a large scale in a cost-effective, non-toxic and eco-friendly way.⁶ Compared with prokaryotic bio-hybridization platforms (e.g., *Escherichia coli*), the transgenic silkworm has a much more versatile capacity to synthesize complex POI by (1) assembling subunit complexes of POI; (2) avoiding the formation of POI inclusion body; and (3) enabling essential post-translational modifications of POI, such as signal peptide excision, phosphorylation, glycosylation, and disulfide bond formation.⁶ Moreover, the transgenic silkworm does not produce endotoxins⁷ and is able to hybridize POI into natural silk with long repeat sequences and high molecular weight. Those advantages are difficult to achieve by bioengineered fermentation.⁸

Progress and potential

Bio-synthesis of recombinant proteins using transgenic silkworms provides a promising bottom-up approach to design and functionalize silk materials. To ensure successful genetic hybridization, protein(s) of interest (POI) are routinely fused with additive domains from natural silk to form fusion POI (F-POI). However, the additive domains often lead to superfluous post-translational modifications and abnormal protein folding, thus severely impairing F-POI. Here, a “light-clothing” strategy, i.e., rationally trimming redundant parts of additive domains, is demonstrated to increase the structural similarity between F-POI and native POI, and, therefore, the activity of F-POI. The silk materials functionalized with highly active F-POI, such as growth factors, can be produced further as biomaterials for tissue regeneration. This strategy provides a solution to overcome the major technical hurdle of using transgenic silkworms as a bio-synthesis platform to develop high-performance materials.



Since the transgenic silkworm technique was developed in 2003,³ many studies have used this advanced bio-synthesis platform to genetically engineer silk-based materials (Table S1). Essentially, the POI synthesized by transgenic silkworms are fusion proteins, i.e., fusion POI (F-POI). As a conventional strategy, additional structural domains derived from endogenic natural silk proteins are routinely introduced upstream and/or downstream of the structural sequences of native POI to form an F-POI. In this way, synthesis and secretion of F-POI and natural silk into cocoons proceeds simultaneously.^{3,4} However, specific parts of the additional structural domains can lead to superfluous post-translational modifications of F-POI and compromise its normal folding.^{4,5,9,10} These abnormalities have been demonstrated to often result in the very low (e.g., 0.9% of the commercially available POI)¹¹ or no activity^{4,12} of F-POI, which seriously limits the practical applications of this advanced bio-synthesis platform for developing high-performance silk-based materials, in particular in the biomedical field.¹³ Therefore, this study aimed to enhance F-POI activity via rationally trimming redundant parts of the additional structural domains without compromising the genetic hybridization process (i.e., synthesis, secretion, extraction, and fabrication) into silk-based materials (Figure 1A). We termed this trimming process the “light-clothing” strategy as the trimmed F-POI is not over-encumbered with additional structural domains, whereas the conventional (i.e., untrimmed) synthesis for F-POI was given the term “heavy-clothing” strategy.

As a proof of concept, transgenic silkworms were used to genetically hybridize growth factors as functional POI into silk fibroin (SF)-based materials for tissue engineering applications by using this light-clothing strategy. SF, the most commonly used silk protein extracted from *B. mori* cocoons, was selected since this protein attracts a mounting interest devoted to tissue engineering applications due to its excellent biocompatibility, robust mechanical properties, hypoallergenic features, and tunable biodegradation.^{14–17} In this study, we first used enhanced green fluorescent protein (GFP) as a marker POI for visualization purposes. By trimming specific redundant parts of the additional structural domains of F-POI, the specific effects of this trimming on the synthesis, secretion, and extraction process of F-POI were investigated in detail. Subsequently, by selecting the human epidermal growth factor (EGF) as a functional model POI, we demonstrated the ease of fabricating a mixture of SF and F-POI into various tissue engineering scaffolds. The activity of this “lightly clothed” F-POI was compared with “heavily clothed” F-POI and native POI (i.e., a commercially available control). Furthermore, the underlying mechanism of the activity of lightly clothed F-POI was studied by theoretical modeling.

RESULTS AND DISCUSSION

Establishment of light-clothing strategy based on trimming redundant parts of additive domains of F-POI

Among various transgenic silk gland expression systems, SF heavy chain (SF-h) expression system was selected in this study to synthesize F-POI in view of its superior efficiency.^{9,18} The SF-h consists of three domains: an N-terminal domain (146 amino acids), a repetitive domain, and a C-terminal domain (57 amino acids) (Figure 1B). The conventional SF-h expression system aims to synthesize a heavily clothed F-POI by replacing the whole repetitive domain of SF-h with the original structural sequences of POI, where additional structural domains derived from SF-h (i.e., N- and C-terminal domains) are kept and fused (Figure 1B).

SF-h is synthesized and secreted by the silk gland of the silkworm.¹⁹ A signal peptide, located on the N terminus, is essential for SF-h to allow for translocation across endoplasmic reticulum membrane, transportation from the silk gland cells into the

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<https://doi.org/10.1016/j.matt.2021.03.020>

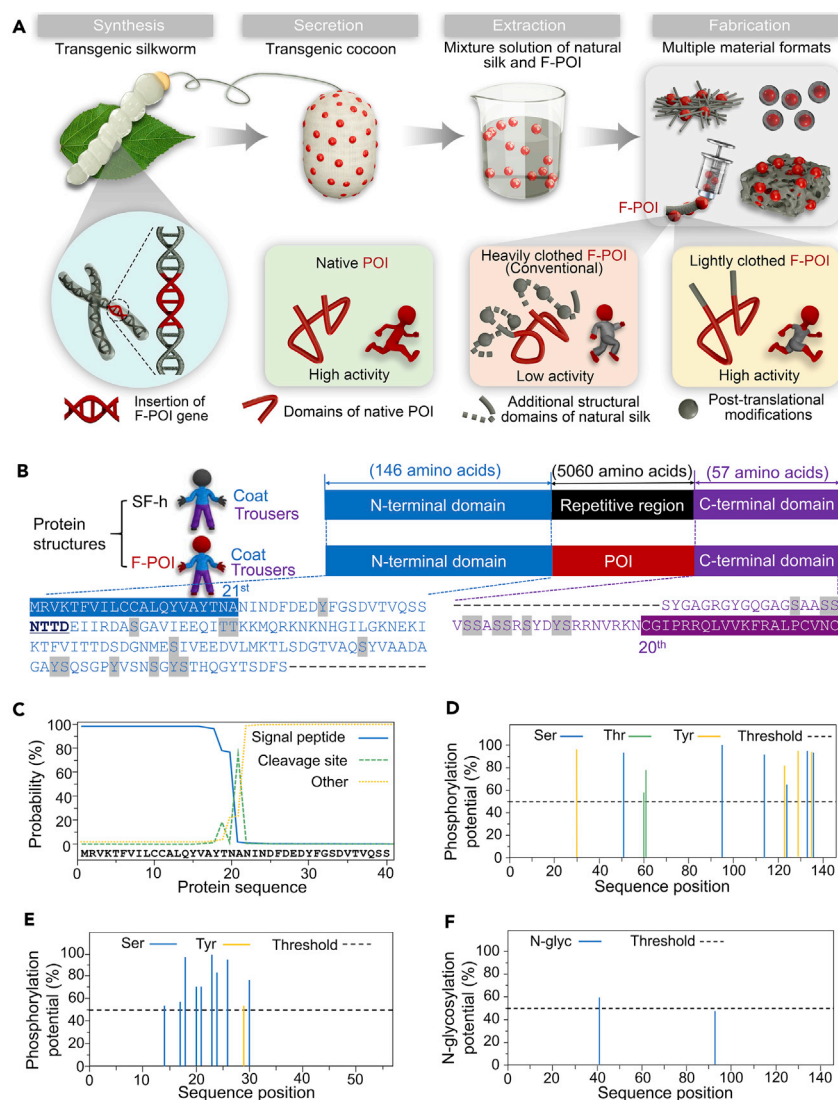


Figure 1. Establishment of light-clothing strategy based on trimming redundant parts of additive domains of F-POI

(A) Genetic hybridization process of F-POI into silk-based materials using conventional or light-clothing strategy.

(B) Conventional SF-h expression systems to synthesize a heavily clothed F-POI and the amino acid sequences of N- and C-terminal domains of F-POI. Gray background and bold blue font correspond to predicted phosphorylation and N-glycosylation sites, respectively. Blue background corresponds to the predicted signal peptide of F-POI. Purple background corresponds to the reported amino acid residues for SF-h folding and assembling.

(C) Prediction of potential signal peptide of F-POI.

(D and E) Prediction of potential phosphorylation sites on N- (D) and C-terminal domains (E) of F-POI.

(F) Prediction of potential N-glycosylation sites on N-terminal domain of F-POI.

lumen, and finally cleavage from the mature SF-h.^{9,20} The signal peptide of heavily clothed F-POI was predicted by SignalP 5.0 software. The results showed that the 1st–21st amino acid residues on the additional N-terminal domain might act as its signal peptide (probability: 98.3%), and the relevant cleavage site of signal peptide might occur between the 21st and 22nd amino acids (probability: 75.5%) (Figures 1B

and 1C; [Table S2](#)). Thus, we speculated that the 1st–21st amino acid residues on the additional N-terminal domain of heavily clothed F-POI (i.e., a shortened N-terminal domain) should be reserved in our strategy, although it could be cleaved during secretion.

Moreover, a recent study found that the last of the 1st–20th amino acid residues on the C terminus of SF-h contains three crucial cysteine residues (1st, 4th, and 20th) ([Figure 1B](#)). These residues can form intra- or intermolecular disulfide bonds, thereby enabling correct folding of natural SF-h and assembly into fibroin units for secretion.²¹ Therefore, we speculated that the last of the 1st–20th amino acid residues on additional C-terminal domain of heavily clothed F-POI (i.e., a shortened C-terminal domain) should also be retained in our strategy based on trimming redundant parts of additive domains.

In addition, we predicted potent post-translational modification sites on additional N- and C-terminal domains of heavily clothed F-POI ([Table S2](#)). We found many potential phosphorylation sites and a potential N-glycosylation site, which are all located outside the regions of shortened N- and C-terminal domains ([Figures 1B, 1D–1F, S1, and S2](#)). For bioengineered proteins, supernumerary post-translational modifications during the heavily clothed F-POI synthesis process can reduce their activity.^{4,5,22} Consequently, we hypothesized that, except for speculated shortened N- and C-terminal domains regions, all other parts of additional N- and C-terminal domains of heavily clothed F-POI should be deleted to enhance the activity of F-POI.

Light-clothing strategy prevents superfluous post-translational modifications in F-POI synthesis

First, GFP was used as a marker POI to visualize the effect of our light-clothing strategy based on trimming redundant parts of additive domains on POI synthesis, secretion, and exaction. Five kinds of F-POI (GFP) with different terminal domains were designed and genetically hybridized into natural SF ([Figures 2A and S4–S6](#); see [supplemental experimental procedures](#) for more details): (1) complete N- and C-terminal domains (NGC group); (2) only complete N-terminal domain (NG group); (3) complete N-terminal domain and shortened C-terminal domain (NGC_s group); (4) only shortened C-terminal domain (GC_s group); and (5) shortened N- and C-terminal domains (N_sGC_s group). The corresponding transgenic strains were constructed and identified ([Figure S7, Tables S4 and S5](#); see [supplemental experimental procedures](#) for more details). Red fluorescence emission from DsRed-Express can be observed in the eyes of moths (adults) ([Figures S4 and S8](#)), demonstrating that all designed F-POI (GFP) gene expression cassettes were successfully inserted into the genomes of transgenic silkworms.^{10,18} The cocoons from the NGC, NG, NGC_s, and N_sGC_s groups displayed strong green fluorescence, while wild-type (WT) and GC_s cocoons did not exhibit fluorescence ([Figure 2B](#)). Correspondingly, no F-POI (GFP) bands could be detected from cocoons in the GC_s and WT groups in the SDS-PAGE and western blot ([Figure 2C](#)). These results pointed out that the complete absence of the N-terminal domain could impede secretion of F-POI into cocoons.

Furthermore, estimated from SDS-PAGE and western blot, the actual molecular weight of F-POI (GFP) for the NGC, NGC_s, and NG groups were 57, 53, and 50 kDa, respectively ([Figures 2C and 2D](#)). These values were all approximately 10 kDa larger than their theoretical values. This increase of the actual molecular weight of F-POI (GFP) has also been reported in previous studies using the

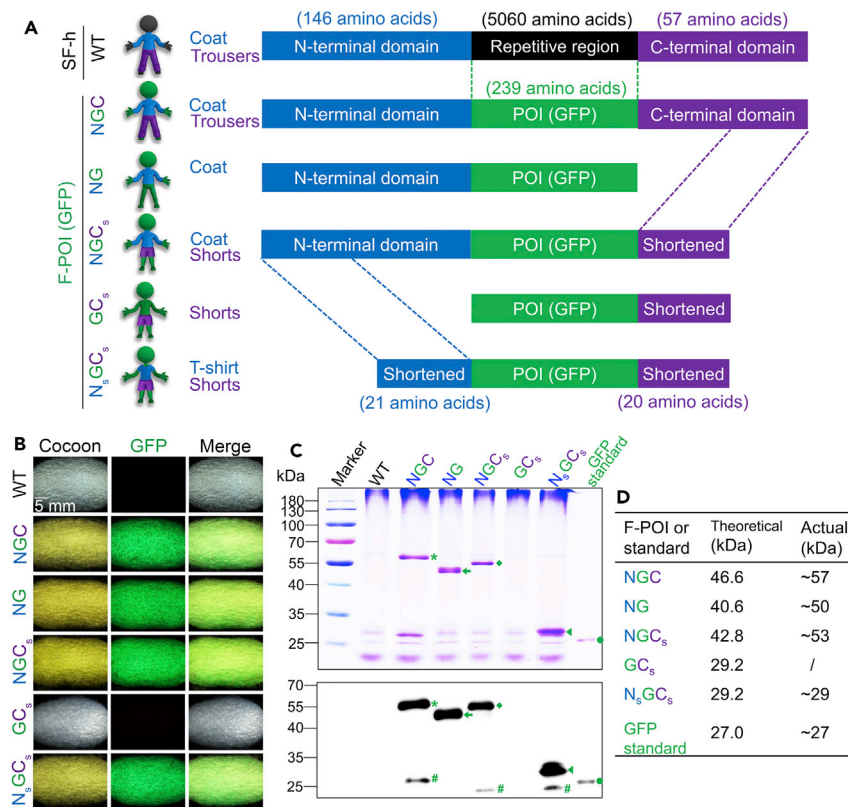


Figure 2. Light-clothing strategy prevents superfluous post-translational modifications in F-POI synthesis

(A) Design of five types of F-POI (GFP) with different terminal domains.
 (B) White light/fluorescent images of cocoons.
 (C) SDS-PAGE (top) and western blot with an anti-GFP antibody (bottom) of the cocoon silks. Green asterisk, arrowhead, rhombus, triangle, or circle indicate the F-POI (GFP) from NGC, NG, NGC_s, and N₅GC_s groups, or commercially available GFP standard, respectively. Green hash indicates the broken protein fragments of F-POI (GFP). The molecular weight values are displayed on the left of the panels.
 (D) Molecular weight of F-POI (GFP) and GFP standard estimated from (C).

conventional SF-h expression system to produce F-POI,^{5,9,10} which can be attributed to the post-translational modifications on additional C- and N-terminal domains of F-POI (Figure 1B). In comparison, the actual molecular weight of F-POI (GFP) in the N₅GC_s group decreased from 53 to 29 kDa as compared with the molecular weight of F-POI (GFP) in the NGC_s group, which was very close to its theoretical value (Figure 2D). This significant decrease indicates that the majority of superfluous post-translational modifications might occur on additional N-terminal domains other than the C-terminal domain. Moreover, the matching of actual and theoretical molecular weights in the N₅GC_s group suggests that superfluous post-translational modifications were successfully prevented by our light-clothing strategy. In addition, the theoretical molecular weights of F-POI (GFP) in the GC_s and N₅GC_s groups were similar (Figure 2D) since the shortened N-terminal domain only contained a predicted signal peptide and would be cleaved during secretion.^{9,20} Another interesting phenomenon is that the bands of broken fragments of F-POI (GFP) were detected in the NGC, NGC_s, and N₅GC_s groups in western blot (Figure 2C), which may result from the cleavage of GFP during the dissolution process of cocoon silks, as reported previously.^{2,10,18}

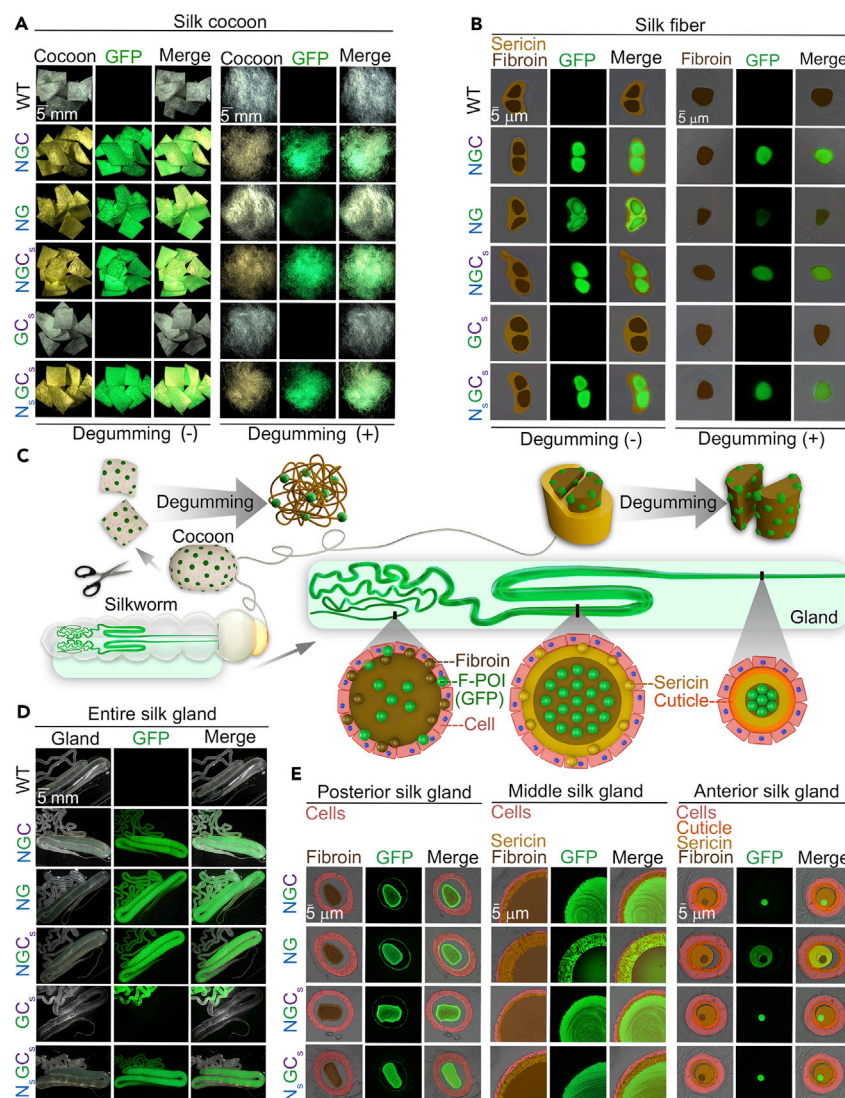


Figure 3. Light-clothing strategy ensures normal secretion and extraction of F-POI

(A and B) White light/fluorescent images of the pieces cut from cocoons (A) and the frozen cross-sections of silk fibers (B).

(C) Schematic illustrating the normal secretion and extraction process of natural silk proteins and F-POI (GFP).

(D) White light/fluorescent images of the entire silk glands.

(E) White light/fluorescent images of frozen cross-sections of the silk glands which have green fluorescence throughout the entire glands in (D). Pseudo colors used to distinguish different anatomical structural regions of silk fibers and the silk glands.

Light-clothing strategy ensures normal secretion and extraction of F-POI

To extract a mixture of SF and F-POI from cocoons for further material preparation, cocoons should be cut into pieces and degummed to remove the outer sericin layer as described previously.⁵ From all four experimental groups exhibiting green fluorescence, only the NG group without a C-terminal domain lost its fluorescence after degumming (Figure 3A). Furthermore, frozen cross-sections of single silk fibers revealed that green fluorescent POI (GFP) was observed in both fibroin and sericin layers in the NG group, while large amounts of F-POI (GFP) were lost after degumming (Figure 3B). Without a complete C-terminal domain, F-POI (GFP) was removed

together with soluble sericin during degumming, which resulted in considerable loss of F-POI (GFP) in the extraction process. Consequently, this group was considered unsuitable for the preparation of SF/F-POI hybrid materials.²³

Furthermore, larval silk glands were dissected to unravel the specific roles of shortened N- and C-terminal domains in F-POI secretion. As shown in Figure 3C, natural SF is first synthesized by posterior silk gland cells and secreted into the lumen. Then, natural SF is transported into middle silk gland, where sericin is synthesized by middle silk gland cells, secreted into the lumen, and wraps on SF. Finally, core (SF)-shell (sericin)-like silk fibers were extruded through the spinneret of silkworms after packing in anterior silk glands.¹⁹ We observed that only the posterior silk gland exhibited green fluorescence in the GC_s group (Figure 3D). This indicates that F-POI without any N-terminal domain can be synthesized by posterior gland cells, but not secreted into the lumen or subsequently transported into middle silk gland.⁹ In contrast, for the other four groups (NGC, NG, NGC_s, and N_sGC_s) with N-terminal domain or shortened N-terminal domain, green fluorescence was observed throughout the entire gland, indicating that these F-POI (GFP) could accomplish the whole secretion process. These results demonstrate that the shortened N-terminal domain designed in our light-clothing strategy could play a similar function by ensuring the progress of F-POI secretion similar to the N-terminal domain in conventional expression systems.

We further prepared frozen cross-sections of silk glands from the four groups with N-terminal domains or shortened N-terminal domains, which were all characterized by complete secretion progress (Figure 3E). F-POI (GFP) in the NG group, i.e., the group without a C-terminal domain, entered the sericin layer in the middle silk gland, and almost all F-POI (GFP) was translocated into the sericin layer in anterior silk glands. After being extruded and spun into cocoons, the majority of F-POI (GFP) still stayed in the sericin layer of silk fibers in the NG group, although some returned to the fibroin layer (Figure 3B). However, the F-POI (GFP) in the other three groups with a C-terminal domain or a shortened C-terminal domain remained in the fibroin layer during the entire secretion process (Figure 3E). Natural SF-h molecules assemble into fibroin units before their secretion into the fibroin layer.^{19,21} We speculate that the retained amino acid residues on the shortened C-terminal domain of F-POI may help F-POI assemble fibroin units with natural SF-h and secrete into the fibroin layer together. F-POI without any C-terminal domain may be in a free state, and can transfer between fibroin and sericin layers during secretion. These results demonstrate that the designed shortened C-terminal domain can prevent the arbitrary transfer of F-POI, similar to the complete C-terminal domain in the conventional expression system. In addition, it should be emphasized that the cocoons of the NG group showed similar fluorescence intensity as that of NGC (Figure 3A), indicating that most of the F-POI could be secreted out without a C-terminal domain.

Genetic hybridization of functional F-POI into various SF-based materials

In the above-described sections, we proved that our trimming of the F-POI using the light-clothing strategy could be used to hybridize F-POI, thereby avoiding supernumerary post-translational modifications and ensuring a normal secretion and extraction. Next, the efficacy of our strategy to enhance the activity of functional F-POI was demonstrated by genetic hybridization of a growth factor into SF-based materials. EGF, a potent growth factor widely used for wound healing,²⁴ was selected as a functional model POI. A lightly clothed F-POI (EGF) with shortened N- and C-terminal domains (N_sEC_s group) and a heavily clothed F-POI (EGF) with complete N- and C-terminal domains (NEC group) were designed and genetically hybridized into

natural SF (Figures S9–S12; see supplemental experimental procedures for more details). Two corresponding transgenic strains were constructed and identified (Figures S13 and S14, Tables S4 and S5; see supplemental experimental procedures for more details).

Red fluorescence was detected in the eyes of moths from the N_sEC_s and NEC groups (Figures S8 and S15), indicating that both F-POI (EGF) gene expression cassettes were successfully inserted into the host genomes.^{10,18} The fluorescent images of silk fiber cross-sections confirmed that functional F-POI (EGF) could be completely and specifically secreted in the fibroin layer, and subsequently survive after the SF/F-POI extraction process (Figure 4A). SDS-PAGE and western blot further confirmed the presence of both intact F-POI (EGF) in transgenic cocoons and regenerated mixture aqueous solutions (Figure 4B). The actual molecular weight of F-POI (EGF) in the NEC group was about 10 kDa larger than its theoretical value, while the actual and theoretical molecular weights of F-POI in the N_sEC_s group were similar (Figures 4B and 4C). These results indicate that redundant post-translational modifications could be avoided for EGF using our strategy.

Quantification of the western blot results showed that the weight ratios of F-POI (EGF) in cocoons were 14.28% and 2.75% for the NEC and N_sEC_s groups, respectively (Figures 4D and S16). Considering the difference of the actual molecular weights (Figure 4D), the expression amount (mole number) of F-POI (EGF) in the NEC group was 30% higher than that in the N_sEC_s group. This difference on molecular amount is determined by different random integration sites of F-POI gene expression cassettes in the silkworm genome (see supplemental experimental procedures for more details). The F-POI expression amount can be further enhanced by remobilizing transposons in the silkworm genome and screening transgenic strains.^{18,25} After extraction, approximately half of F-POI in both groups were retained in regenerated mixture aqueous solutions (Figure 4D). For the N_sEC_s group, the elasticity, tensile strength, Young's modulus, and toughness of degummed silk fibers were lower than those of WT while higher than those of the NEC group, but not statistically significant (Figures 4E–4H and S17). We demonstrated the feasibility of using this hybrid aqueous solution to fabricate commonly used material formats for tissue engineering applications similar to natural SF, such as injectable hydrogels, foam scaffolds, casting films, nanofiber meshes, and nanospheres (Figure 4I; see experimental procedures for more details). Therefore, we anticipate that this ready-to-use mixture solution of SF and F-POI will become a versatile toolbox for fabricating various downstream SF-based materials similar to natural SF.²³ One thing to note here is that the properties of these SF-based materials can be easily adjusted by changing the preparation parameters (e.g., pore sizes and annealing process) to meet different requirements for various applications (e.g., soft or hard tissue regeneration).²³

Activity of functional F-POI hybridized in SF-based materials

Finally, SF films, as a material format model, were prepared to compare the activity of two types of F-POI hybridized into SF-based materials (Figure S18). NIH 3T3, a mouse fibroblasts cell line, was selected to measure the activity of EGF as described previously.²⁴ Cells were cultured in the presence of extracts from NEC and N_sEC_s films as well as an aqueous solution of commercially available EGF standard (positive control group), which was diluted to a concentration of 0.25 nM for optimal stimulation of NIH 3T3 cells by EGF (Figure S19; see experimental procedures for more details). Since the diluted extracts from WT SF films had no significant influence on cell proliferation (Figure S20), cell culture medium without EGF was used as a negative control group.

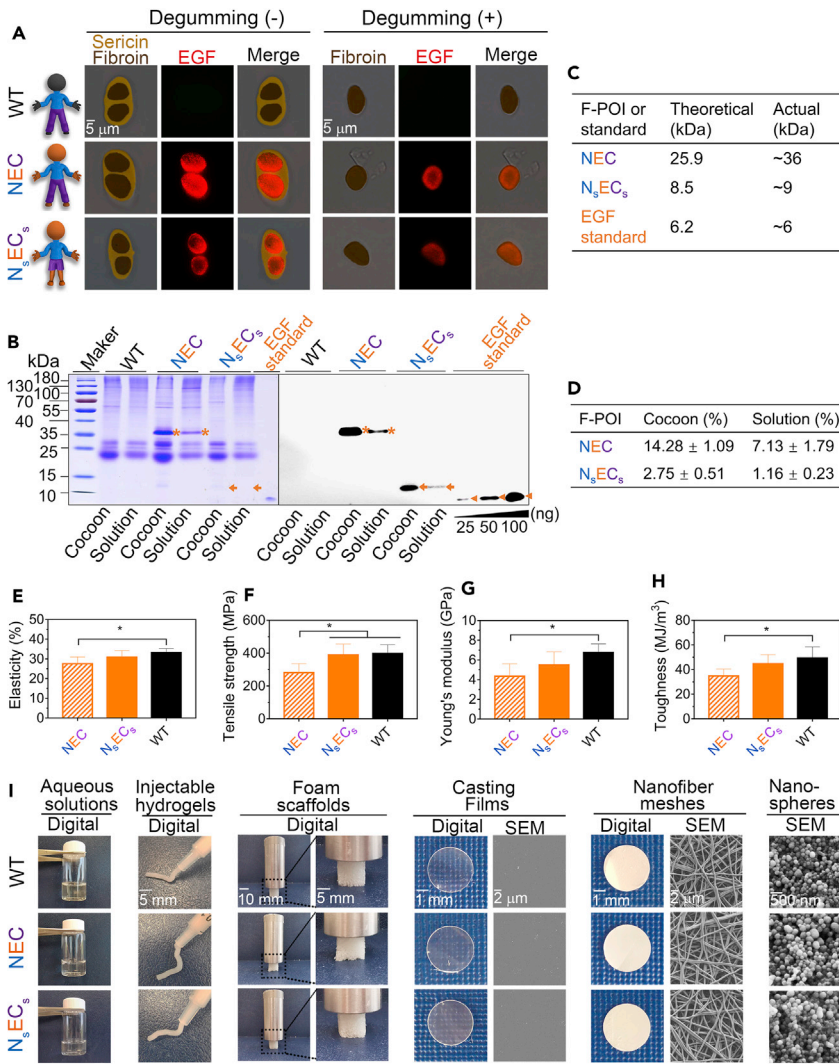


Figure 4. Genetic hybridization of functional F-POI into various SF-based materials

(A) White light/fluorescent images of frozen cross-sections of silk fibers. Pseudo colors used to distinguish different anatomical structural regions of silk fibers.

(B) SDS-PAGE (left) and western blot with an anti-EGF antibody (right) of the cocoons and the regenerated aqueous solutions. Orange asterisk, arrowhead, or triangle indicate the F-POI (EGF) in NEC and N_sEC_s groups, or commercially available EGF standard, respectively. The molecular weight values are displayed on the left of the panels.

(C) Molecular weight of F-POI (EGF) and EGF standard evaluated from (B).

(D) Weight ratio of F-POI (EGF) to total proteins in different groups by quantitative analysis of western blot results. Data are represented as mean ± SEM.

(E–H) Mechanical properties of degummed silk fibers from WT, NEC, and N_sEC_s groups. Elasticity (E), tensile strength (F), Young's modulus (G), and toughness (H). Error bars representing one standard deviation (*p < 0.05, **p < 0.01).

(I) Digital photos and SEM images showing the feasibility of fabricating injectable hydrogels, foam scaffolds (a weight of 200 g put on them), casting films, nanofiber meshes, and nanospheres from regenerated aqueous solutions.

Ki-67 protein is a proliferation marker associated with ribosomal RNA transcription, which is present during all the active phases (G1, S, G2, M) of the cell cycle.^{26,27} After 24 h, more Ki-67-positive cells were found for the N_sEC_s and EGF standard groups than the NEC (p < 0.05) and negative control groups (p < 0.01) (Figures

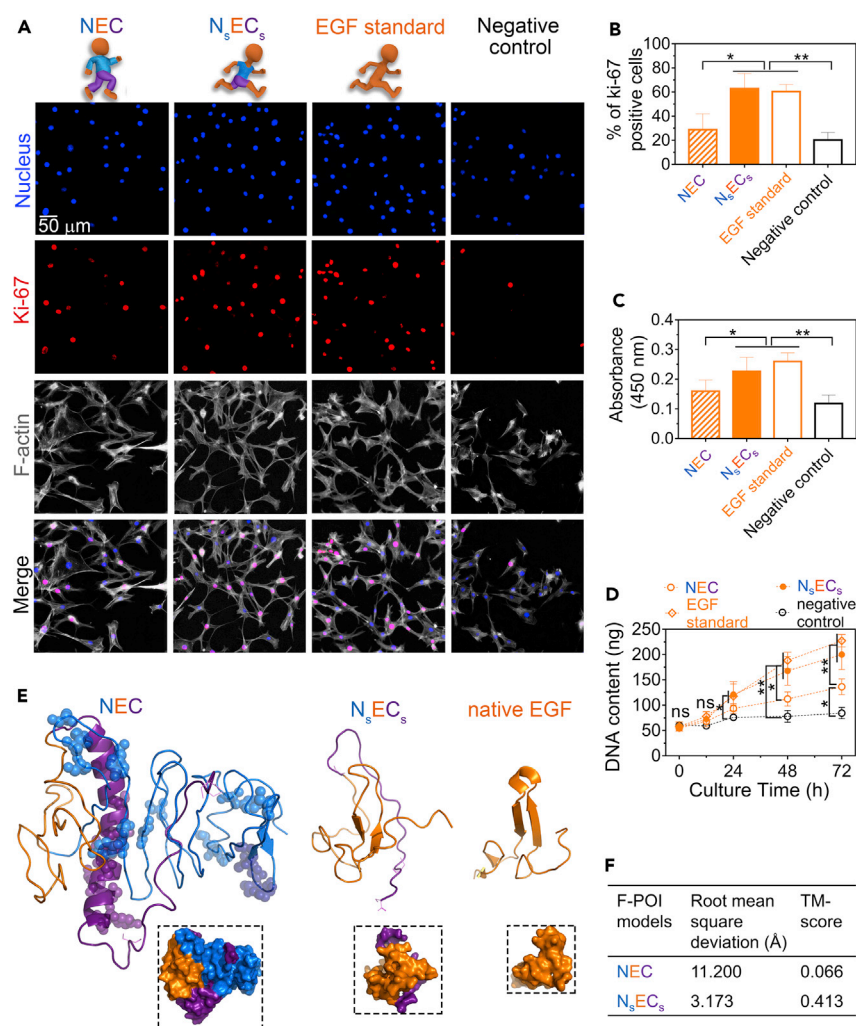


Figure 5. Activity of functional F-POI hybridized in SF-based materials

(A) Representative immunofluorescent images of nucleus (blue), Ki-67 protein (red), and F-actin (gray) in cells at 24 h.

(B) Cell proliferation state at 24 h measured by Ki-67 staining.

(C) Cell metabolism activity at 72 h measured using a CCK-8 kit.

(D) Cell proliferation rate within 72 h measured using a QuantiFluor dsDNA kit.

(E) *De novo* protein modeling of the F-POI (EGF) and the native EGF model. Purple, orange, and blue bands indicate the N-terminal domain, the structural segment of native EGF, and the C-terminal domain (or shortened C-terminal domain), respectively. Blue and purple spheres indicate the predicted phosphorylation sites on N- and C-terminal domains, respectively. Dark blue spheres indicate the predicted glycosylation sites on the N-terminal domain. Pink lines indicate the crucial cysteine residues on the C-terminal domain (or shortened C-terminal domain). The corresponding surface models are displayed in black dashed line frames.

(F) C-alpha atomic root-mean-square deviation values and TM scores of the F-POI (EGF) models versus the native EGF model. Error bars representing one standard deviation (ns, not significant; * $p < 0.05$, ** $p < 0.01$).

5A and 5B). This result indicates that the activity of lightly clothed F-POI (EGF) is similar to the commercially available EGF standard by activating more cells into proliferation than the heavily clothed F-POI (EGF). After 72 h, Cell Counting Kit-8 (CCK-8) test showed that lightly clothed F-POI (EGF) and commercially available EGF standard significantly improved the metabolic activity of cells as compared

with heavily clothed F-POI (EGF) ($p < 0.05$) (Figure 5C), which were in line with the Ki-67 assay.

Moreover, DNA contents were measured to quantify cell proliferation (Figure 5D). Within the first 12 h, proliferation was similar for all four experimental groups. After 24 h, cell numbers were significantly higher in the positive control and N_sEC_s groups than that in the negative control group ($p < 0.05$). The difference became larger with prolonged culturing time. After 72 h, cell proliferation in the presence of positive control and N_sEC_s groups was significantly higher than that in the NEC group ($p < 0.01$), while cell number in the NEC group was significantly higher than that in the negative control group ($p < 0.05$). Taken together, these results indicate that the activity of heavily clothed F-POI (EGF) was much lower than lightly clothed F-POI (EGF). Furthermore, the activity of lightly clothed F-POI (EGF) was comparable with the commercially available EGF standard.

Biologically, protein function is determined by its three-dimensional (3D) structure.²⁸ To evaluate the structural similarities of the F-POI to native POI, *de novo* protein modeling was therefore performed to build the 3D structures of lightly and heavily clothed F-POI (EGF) using a deep learning-based approach, DMPfold (the DMPfold server: <http://bioinf.cs.ucl.ac.uk/psipred/>)²⁹ (Figures 5E and S21). The lightly clothed F-POI (EGF) and native EGF had a similar center-aligned β sheet, as opposed to the heavily clothed F-POI (EGF). As shown in Figure 5F, the C-alpha atomic root-mean-square deviation value between lightly clothed F-POI (EGF) and native EGF (3.173 Å) is much smaller than that between the heavily clothed F-POI (EGF) and native EGF (11.200 Å). Furthermore, the predicted structure of lightly clothed F-POI (EGF) has a high TM score of 0.413 to native EGF, while that of the heavily clothed F-POI (EGF) yielding a TM score of 0.066 to native EGF is considered random (i.e., the TM score ranging from 0 to 0.17).³⁰ In combination, these data indicate that our light-clothing strategy improved the structural similarity between F-POI (EGF) and native EGF, which might also contribute to the enhanced activity of functional F-POI (EGF) besides the removal of supernumerary post-translational modifications.

Furthermore, when EGF was replaced by other reported functional F-POI hybridized by transgenic silkworms, such as human basic fibroblast growth factor 2 (FGF-2)¹¹ and human insulin-like growth factor I (IGF-1),³¹ the lightly clothed F-POI still had a much higher structural similarity to native POI than the heavily clothed F-POI, with the removal of redundant post-translational modifications (Figures S22 and S23). We speculate that our light-clothing strategy might also be applicable to other functional F-POI to enhance their activity.

Compared with direct incorporation of commercially available POI into silk-based materials, these genetically hybridized silk-based materials not only have comparable POI activity, but also provide additional advantages. First, cold chain logistics and storage are not necessary for hybridizing (sensitive) POI. The transgenic cocoons in this study were shipped and stored at room temperature for more than half a year before used, whereas commercially available EGF standard can be only shipped and stored at -20°C . Previous studies also reported that silk could stabilize and retain the activity of viruses, blood samples, and biomacromolecules (e.g., antibiotics and antibodies) at room/elevated temperature for months via encapsulation.³²

Moreover, genetic hybridization of SF with POI are almost free of charge, especially for expensive POI, since the transgenic silkworm platform produces natural SF and

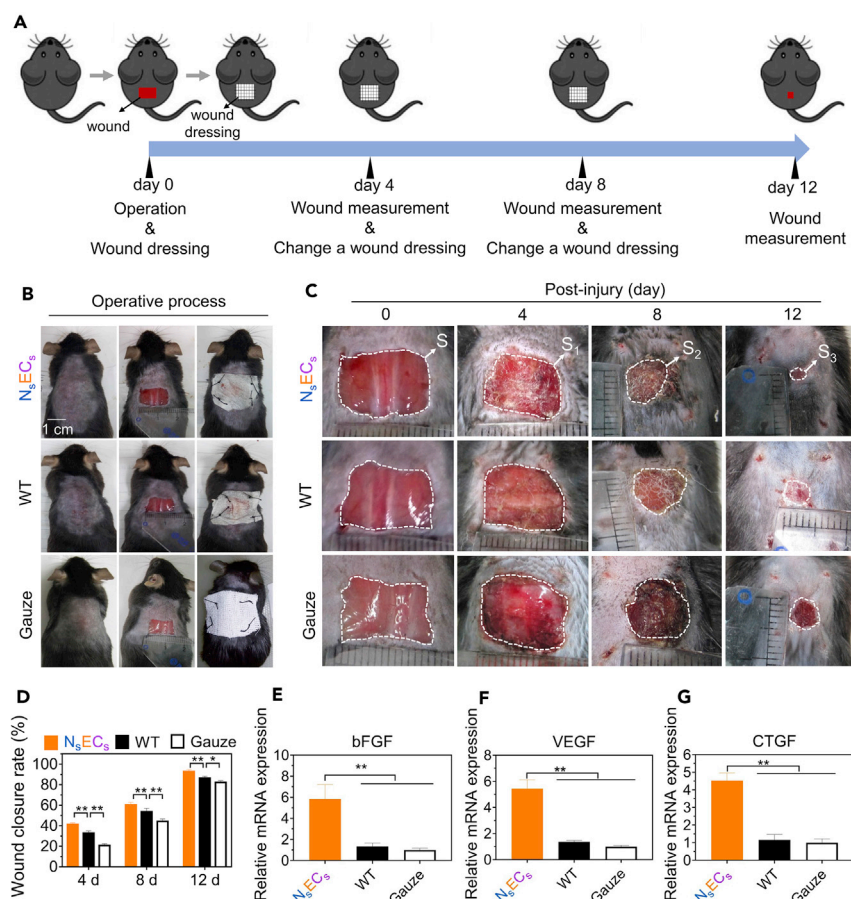


Figure 6. In vivo assessment of the genetically hybridized silk films for healing mouse excisional wounds

(A) Schematic of the time course of the mouse excisional skin wound model.
 (B) Operative process for creating wound models on day 0.
 (C) Comparison of wound closure among the N_sEC_s , WT, and gauze groups. Representative photographs of wounds on days 0, 4, 8, and 12 post-injury are shown. Each grid of the ruler is 1 mm. The wound on day 0 is from the enlarged image of the corresponding wound in (B). All the images from each group shown the progression of wound healing in the same animal.
 (D) Quantitative evaluation of the wound area closure on days 4, 8, and 12 post-injury.
 (E–G) mRNA expressions of fibroblast growth factor 2 (bFGF) (E), vascular endothelial growth factor A (VEGF) (F), and connective tissue growth factor (CTGF) (G) from the wound tissue. Error bars representing standard deviations (*p < 0.05, **p < 0.01).

POI simultaneously without extra POI purchase cost. However, it has to be noted that the extraction of SF with POI, as well as WT SF, involves the use of some inexpensive reagents, such as lithium bromide. Furthermore, when the desired application does not require the presence of SF, purification of POI away from SF would add some extra cost to the production of POI with this system. In addition, POI in this study, i.e., EGF, is relatively small and refolded well after routine SF/F-POI extraction procedure. However, refolding a large and complex F-POI after routine extraction procedure may be a challenge and raise the cost of using this bio-synthesis system. One possible solution is to use alternative extraction methods, such as directly extracting SF/F-POI from dissected silk glands¹² to avoid F-POI refolding process.

Finally, to demonstrate the potential biomedical applications of our developed genetically hybridized silk material system, we used films made of WT and N_sEC_s

SF to heal skin wounds in a mouse excisional skin wound model (Figures 6A and 6B). By measuring the wound closure rate (Figures 6C and 6D) and the expression levels of wound healing-associated genes in the affected skin tissues (Figures 6E–6G), we found that wound healing process was significantly accelerated by using the lightly clothed silk as compared with the WT silk and a commercial wound dressing. This beneficial effect can be attributed to the genetically hybridized high active EGF in the N₅EC_s silk films.

Although we only focused on silk in this study, other natural proteins, such as collagen and elastin, can also be bio-synthesized by transgenic animals and plants.^{3,33} These natural protein-based materials are extensively studied in various application areas.^{34,35} We envision that our developed light-clothing strategy can also be exploited for genetic hybridization of highly active functional F-POI into these natural protein-based materials to meet application-specific needs.

Conclusions

In summary, we proposed a strategy, termed light-clothing, based on trimming the redundant additive structural domains for genetic hybridization of highly active functional fusion POI into silk-based materials using transgenic silkworm-based bio-synthesis platforms. As compared with the conventional expression systems, our strategy remarkably enhanced the activity of fusion POI, without compromising its genetic hybridization process. This enhanced activity of fusion POI can be attributed to the removal of supernumerary post-translational modifications and increased structural similarity to native POI. This study provides fundamental insights and a feasible solution to overcome the major technical hurdle, i.e., seriously limited activity of fusion POI, in utilizing these advanced bio-synthesis platforms to develop high-performance silk-based materials.

EXPERIMENTAL PROCEDURES

Resource availability

Lead contact

Further information and requests for resources and reagents should be directed to and will be fulfilled by the lead contact, Aichun Zhao (zhaoaichun@hotmail.com).

Materials availability

Plasmids and transgenic cocoons generated in this study will be made available on request, but we may require a payment and/or a completed Materials Transfer Agreement if there is potential for commercial application.

Data and code availability

All data needed to evaluate the conclusions in the paper are presented in the paper and/or supplemental information. Original data have been deposited to Mendeley data: <https://data.mendeley.com/datasets/r8y38sjwgv>. Additional data related to this research may be requested from the authors.

Silkworm strain

The diapause *B. mori* strain 871 (commercial strain, white cocoon) was maintained at the Silkworm Resource Bank of Southwest University, Chongqing, China. The diapause character of eggs was changed by using the 15°C-IMES (incubating mother eggs at 15°C) strategy.³⁶ Non-diapause eggs after DNA pre-blastoderm microinjection were maintained at a temperature of 25°C and relative humidity of 85%–90%, and hatched larvae were reared at a temperature of 25°C and relative humidity of 75%–80% and fed mulberry leaves.

Predictive analysis of signal peptide and post-translational modification sites

Sequences of the N terminus (146 amino acids) and the C terminus (57 amino acids) of heavily clothed F-POI were sent to online programs (Table S2) predicting for signal peptide sequence and the potential post-translational modification sites (thresholds set to 50%).

Design and construction of piggyBac-derived vectors

The codon sequences of 2,304-bp FHP3, 895-bp FHP1, 1,929-bp FHP3_s, 333-bp LBS, 222-bp LBS_s, and 162-bp poly(A) derived from the SF heavy chain gene of *B. mori* (NCBI Gene ID: 693030), the codon sequence of 717-bp GFP derived from enhanced GFP gene (NCBI Reference Sequence: NC_025025.1), and the optimized codon sequence of 159-bp EGF with a silkworm codon bias derived from the human EGF (NCBI Reference Sequence: NM_001963.6) (Figure S3), were synthesized commercially (GenScript, Nanjing, China). The above fragments were combined and seamlessly inserted into the pUC57-T-simple vector to generate plasmids pUC-FHP3-GFP-LBS, pUC-FHP3-GFP-poly(A), pUC-FHP3-GFP-LBS_s, pUC-FHP1-GFP-LBS_s, pUC-FHP3_s-GFP-LBS_s, pUC-FHP3-EGF-LBS, and pUC-FHP3_s-EGF-LBS_s, respectively, which carried *Ascl* and *FseI* restriction sites (GenScript). The piggyBac-derived vectors pBac{3×P3-DsRed; FHP3-GFP-LBS}, pBac{3×P3-DsRed; FHP3-GFP-poly(A)}, pBac{3×P3-DsRed; FHP3-GFP-LBS_s}, pBac{3×P3-DsRed; FHP1-GFP-LBS_s}, pBac{3×P3-DsRed; FHP3_s-GFP-LBS_s}, pBac{3×P3-DsRed; FHP3-EGF-LBS}, and pBac{3×P3-DsRed; FHP3_s-EGF-LBS_s} were constructed by cloning the *Ascl*/*FseI* fragments from pUC-FHP3-GFP-LBS, pUC-FHP3-GFP-poly(A), pUC-FHP3-GFP-LBS_s, pUC-FHP1-GFP-LBS_s, pUC-FHP3_s-GFP-LBS_s, pUC-FHP3-EGF-LBS, and pUC-FHP3_s-EGF-LBS_s into *Ascl*-cut pBac{3×P3-DsRed},³⁷ respectively. All DNA fragments and resulting piggyBac-derived vectors were confirmed by DNA sequencing.

Generation, screening, and molecular identification of transgenic silkworms

Germline transformation of *B. mori* was carried out using a previously described piggyBac transposon-based transgenic method.^{38–40} The copy number and chromosomal insertion sites of the transgene constructs were determined by Southern blot and inverse PCR, respectively, as described previously with some improvements.^{38,39,41} Full experimental procedures are provided in the supplemental information.

Extraction of a mixture of SF and F-POI from silk cocoons

To remove the sericin, degumming of the cocoons was carried out using a method reported previously.⁵ The cocoons were first cut into small pieces and degummed in a mixture of 0.08% (w v^{−1}) sodium carbonate and 0.12% (w v^{−1}) Marseille soap at 70°C for 45 min and then rinsing several times with distilled water. Regenerated mixture aqueous solutions of SF and F-POI (or natural SF solution) were prepared from the degummed silk fibers as reported previously.²³ In brief, the degummed silk fibers were dissolved in 9.3 M LiBr solution for 2.5 h at 60°C before dialyzing in dialysis membranes (molecular weight cutoff: 3,500, Pierce) against water to remove the salt. Finally, the regenerated aqueous solutions were centrifuged to remove the impurities and stored at 4°C. Their concentrations were calculated by a dry weighing method.

SDS-PAGE and western blot

The sample preparation and treatment protocols of the SDS-PAGE and western blot analysis were described previously.^{10,18} About 30 mg of each silkworm cocoon without or with degumming was dissolved in 500 μL of 9.3 M LiBr solution for

2.5 h at 60°C. Protein concentrations of the mixture aqueous solutions collected after dialysis were detected with the bicinchoninic acid protein assay kit (Beyotime Biotech, Taicang, China). The commercially available GFP standard (ab84191, Abcam, USA) or EGF standard (AF-100-15, PeproTech, USA) was used as a control. The regenerated mixture aqueous solutions or control was mixed with 2% ($v\ v^{-1}$) β -mercaptoethanol and boiled for 5 min, and then subjected to SDS-PAGE (15%, Bio-Rad). The gel was visualized by 0.1% ($w\ v^{-1}$) Coomassie brilliant blue R-250, 50% ($v\ v^{-1}$) methanol, and 10% ($v\ v^{-1}$) acetic acid.

Western blot was performed using anti-GFP antibody or anti-EGF antibody and its corresponding secondary antibodies (Table S6). Immunoreactive protein bands were visualized using the chemiluminescence ECL-plus kit (Beyotime Biotech, Jiangsu, China), and detected by using a chemiluminescence imaging system (ChemiScope 5300; Clinx Science Instruments, Shanghai, China). The intensities of the resulting bands were calculated with ImageJ software (<https://imagej.nih.gov/ij/>) and correspond to the mean $\pm$ standard deviation of three independent experiments.

Analysis of F-POI in transgenic cocoons, fibers, and silk glands

Cocoons from transgenic silkworms were randomly selected and cut into pieces, after which the entire cocoons and their cut pieces before or after degumming were photographed using the fluorescence stereomicroscope equipped with a GFP filter (Olympus MacroViewMVX10-AUTO, Tokyo, Japan).

The transgenic silk fibers without or with degumming were twisted into a bundle and embedded into Tissue-Tek OCT compound (Sakura Finetek, Torrance, CA, USA), and then frozen at -40°C overnight. Subsequently, 5- μm -thick slices were observed under a fluorescence microscope (BX51TRF Olympus, Tokyo, Japan).

Samples of transgenic silk fibers for histological examination were prepared as described previously.⁴² In brief, the transgenic silk fibers without or with degumming were fixed overnight with 10% ($v\ v^{-1}$) formalin, frozen in Tissue-Tek OCT compound, and cut into 10- μm -thick sections. Sections were immunohistochemically stained using anti-EGF antibody and its corresponding secondary antibody (Table S6) and observed under the fluorescence microscope (BX51TRF Olympus).

Silk glands of day 5 fifth-instar larvae were dissected, and the entire silk glands placed on glass slides and observed under the fluorescence stereomicroscope equipped with a GFP filter (Olympus MacroViewMVX10-AUTO). Silk glands were cut into small segments and immersed in Tissue-Tek OCT compound and frozen at -40°C overnight. Then the 7- μm -thick sections of the posterior, middle, and anterior silk glands were observed under the fluorescence microscope (BX51TRF Olympus), respectively.

Mechanical testing of silk fibers

Five cocoons degummed as above were randomly selected from each group. Mechanical properties of each cocoon were calculated by the average of the specimens of ten different silk fibers randomly isolated from each cocoon. The tensile testing of silk fibers was performed using a Universal Material Testing Machine (Instron 5567, MN, USA) at room temperature with a gauge length of 10 mm and a strain rate of 10 mm min^{-1} . The mechanical data (load and elongation) were recorded with TRAPEZIUM 1.26 software (Shimadzu) until the fiber broke, which permitted the calculation of stress and strain values. Stress-strain curves and average curves by curve

fitting were calculated and drawn using ORIGIN 8.0 (OriginLab, Northampton, MA, USA) software. Elasticity, tensile strength, Young's modulus, and toughness of the silk fiber sample were calculated also using ORIGIN 8.0. software according to the method reported previously.⁴³

Feasibility of fabricating various SF-based materials

According to the weight ratio calculated from western blot, the weight ratio of all F-POI was diluted to 1 wt% of total proteins using natural aqueous SF solution. Hydrogels were prepared by a commonly used sonication method.⁴⁴ In brief, mixture solutions of SF and F-POI (or natural SF solution) (5 wt%) were applied via an ultrasonic treatment (UP50H, Hielscher, Germany), which lasted for 1 min at 20% amplitude with a pulse rate of 1 s on and 1 s off. Then solutions were poured into syringes and incubated at 4°C overnight. Finally, the sol-gel transition was confirmed visually, and the injection of hydrogels from syringes was observed by a digital camera.

Foam scaffolds were prepared by a casting method described previously.⁴⁵ In brief, mixture solutions of SF and F-POI (or natural SF solution) (2 wt%) were poured into plastic Petri dishes. They were first cooled down to 4°C and transferred to −20°C overnight to induce phase separation. Then, they were stored at −80°C for another 24 h before freeze drying. After drying, scaffolds were annealed by water vapor at room temperature overnight. Finally, they were cut into cubes and observed by digital camera.

Films were prepared by a commonly used casting method for natural SF films.²³ In brief, mixture solutions of SF and F-POI (or natural SF solution) (2 wt%) were cast on plastic Petri dishes. After air drying, the films were annealed by water vapor at room temperature overnight.⁴⁶ Finally, they were punched into discs and observed by scanning electron microscope (Sigma-300, Zeiss, German) and digital camera.

Nanofiber meshes were prepared by a commonly used electrospinning method.⁴⁷ In brief, poly(ethylene oxide) was dissolved in mixture solutions of SF and F-POI (or natural SF solution) (7.5 wt%) with a weight ratio of 0.3. The mixtures were electrospun using an electrospinning machine (Esprayer ES-2000S, Fulence, Japan) at a flow rate of 20 $\mu\text{L min}^{-1}$, a voltage of 20–25 kV, and a distance of 20 cm. After electrospinning, nanofiber meshes were annealed by water vapor at room temperature overnight. Finally, they were punched into discs and observed by SEM and digital camera.

Nanospheres were prepared by a commonly used precipitation method.^{14,48} The mixture solutions of SF and F-POI (4 wt%) or SF solution (4 wt%) were dropwise added into acetone. Then, nanospheres were washed with water and centrifuged three times at 5,000 rpm for 1 h. Finally, the morphology of nanospheres was observed by SEM.

Cell culture

NIH 3T3 (ATCC, USA), a mouse embryonic fibroblastic cell, was cultured in Dulbecco's modified essential medium (DMEM, Gibco, Scotland) with the addition of 10% calf serum (Gibco) in humidified 5% CO₂ at 37°C.

Measurement of optimum molar concentrations of EGF for cultured cells

To measure the optimum molar concentration of EGF in the following research, NIH 3T3 cells were first seeded into a 96-well plate at a density of 10,000 cells cm^{−2}. After culture in DMEM (10% calf serum) for 24 h, cells were starved for another 24 h by replacing the culture medium with DMEM (0.5% calf serum). Then, cells were further

cultured for 72 h in DMEM (0.5% calf serum) containing different concentrations of commercially available EGF standard (AF-100-15, PeproTech). Cell proliferation metabolic activity was tested using a cell-counting kit (CCK-8; Dojindo, Japan). The results were measured using a spectrophotometer (BioTek FL600 microplate fluorescence reader, BioTek, USA) according to the manufacturer's instructions. DMEM containing no EGF was used as a negative control. Three wells per group were tested.

Dilution of released F-POI and commercially available POI standard

UV light was used to sterilize prepared films. Then films were placed in 48-well cell culture plates and immersed in 500 mL DMEM (0% calf serum) for 24 h. The concentrations of two kinds of F-POI (EGF) in collected extracts were investigated using an ELISA kit (900-K05, PeproTech), and absorbance was read by spectrophotometer according to the manufacturer instructions. Then the extracts were diluted to the optimum molecular molar concentration of EGF (with 0.5% calf serum) using culture medium. Commercially available EGF standard was diluted in the same way. Three samples per group were prepared.

Activity measurement of F-POI and commercially available POI standard

NIH 3T3 cells were seeded into a 96-well cell culture plate at a density of 10,000 cells cm^{-2} . After cells were cultured in DMEM (10% calf serum) for 24 h, the culture medium was replaced with DMEM (0.5% calf serum), and cells were starved for 24 h. Then, cells were cultured for 72 h in diluted extracts from NEC and N₃EC_s groups, or commercially available EGF standard. The DMEM (0.5% calf serum) without EGF was used as a negative control (Figure S18).

To observe cell spreading and proliferating at 24 h, Ki-67 and F-actin were stained. In brief, cells were fixed in 4% paraformaldehyde for 10 min and permeabilized with 0.1% Triton X-100. Subsequently, cells were incubated with corresponding primary and secondary antibodies (Table S6) and incubated for 1 h, respectively. Phalloidin and DAPI (Table S6) was applied to stain F-actin and nuclei, respectively. Images were taken using fluorescent microscopy (Axio Imager Microscope Z1, Zeiss, Germany). Three random fields across three samples per group were measured.

Cell proliferation metabolic activity at 72 h was measured using a cell-counting kit (CCK-8; Dojindo, Japan). The result of CCK-8 was tested using spectrophotometer according to the manufacturer instructions. Cell proliferation rate within 72 h was measured via QuantiFluor dsDNA System kit (E2670, Promega Corporation, Madison, WI, USA). The result of DNA content was tested using spectrophotometry according to the manufacturer's instructions.

De novo protein modeling

DMPfold (the available server: <http://bioinf.cs.ucl.ac.uk/psipred/> in which the protein sequences were only required for protein modeling), a recently released deep learning approach for protein modeling,²⁹ was used to build the 3D structures of heavily and lightly clothed F-POI (EGF), respectively, as well as other F-POI (human basic fibroblast growth factor [FGF-2] and human insulin-like growth factor I [IGF-1]). The method produced 3D structures of the F-POI by combining the predicted information of the inter-residue C β distances, H-bonds, and torsion angles using a collection of features based on the F-POI or/and their homologous sequences (i.e., multiple sequence alignments). The amino acid sequences of the F-POI were solely involved in the course of 3D structure modeling. The crystal structures of native

EGF (PDB: 1JL9), FGF-2 (PDB: 1BFG), and IGF-1 (PDB: 1BQT) were derived from the protein databank: <https://www.rcsb.org/>. The template modeling scores (TM scores) and C-alpha atomic root-mean-square deviation values were calculated using the C++ program: <https://zhanglab.ccmb.med.umich.edu/TM-score/>.

In vivo wound healing evaluation

Eight-week-old male C57BL/6J mice (Center of Laboratory Animals of Third Military Medical University, Chongqing, China) were used in this study and the animal experiment was approved by the Ethics Committee of Southwest University. After anesthesia, the back hair of mice was removed, and the wound model was prepared by cutting a 10 × 15 mm incision into back skin. Silkworm cocoons from N₃EC_s or WT groups were degummed using above method and cut into a 20 × 30 mm film as previous studies reported.^{49,50} Films were sterilized by UV light. Each wound was covered with different silk films or gauzes (five mice per group), which were fixed by sutures (Figure 6B). All mice were bred in separated cages to prevent licking or touching the wound tissues of animals. On days 4 and 8, the used silk films or gauzes were carefully dismantled and changed with a new wound dressing after anesthesia. On days 0, 4, 8, and 12, digital wound photographs were taken with a ruler as calibrator. The wound areas were calculated and denoted by S, S₁, S₂, and S₃, respectively. The wound closure rate (%) was calculated by:

$$\text{Wound closure rate} \left(\% \right) = \frac{S - S_x}{S} \times 100 \% \quad (x = 1, 2, \text{ or } 3)$$

Mice were euthanized on day 12 day. Following sacrifice, skin tissue of the wound edge was obtained and immediately stored in the RNeasy RNA Stabilization Reagent (QIAGEN, Valencia, CA, USA) at 4°C overnight and frozen at −20°C. Total RNA was extracted from harvested wound tissues using TRIzol Reagent (Invitrogen, Carlsbad, CA, USA). cDNA was synthesized using a PrimeScript RT Reagent Kit with gDNA Eraser (Takara Shuzo, Kyoto, Japan). Real-time quantitative reverse-transcription PCR was performed to measure gene expression using the primers in Table S3. Glyceraldehyde-3-phosphate dehydrogenase (GAPDH) served as an internal control.

Statistics

One-way ANOVA was used to determine statistical significance followed by post hoc analysis using the Tukey test. All statistical analyses were performed with GraphPad prism.

SUPPLEMENTAL INFORMATION

Supplemental information can be found online at <https://doi.org/10.1016/j.matt.2021.03.020>.

ACKNOWLEDGMENTS

The authors thank Mr. Jinmeng Li for his valuable suggestions, Jairo Cordeiro for SEM images, and Dr. Brandon Scott Canup for language editing. We also would like to express our sincere gratitude to Prof. Sander C.G. Leeuwenburgh and Prof. John A. Jansen for their valuable comments and advice during this study. This work was supported by the National Natural Science Foundation of China (31801126), the China Agriculture Research System (CARS-18-ZJ0201), the China Scholarship Council (201606160095), and the Doctoral Start-up Foundation of Southwest University (SWU120010). We also thank the Erasmus+ staff mobility for teaching and training between PROGRAMME and PARTNER COUNTRIES

(NIJMEGE01) for providing support for the exchange program at Radboud University, Nijmegen, the Netherlands.

AUTHOR CONTRIBUTIONS

Conceptualization, D.L. and X.C.; methodology, D.L., X.C., and J.S.; data curation, D.L., X.C., Z.H., J.S., D.U., Y.L., and L.C.; investigation, D.L., X.C., and Z.H.; software, J.S.; supervision, Z.X., F.Y., and A.Z.; funding acquisition, D.L., X.C., and A.Z.; writing – original draft, D.L. and X.C.; writing – review & editing, D.L., X.C., F.Y., and A.Z.

DECLARATION OF INTERESTS

The authors declare no competing interests.

Received: December 17, 2020

Revised: February 21, 2021

Accepted: March 24, 2021

Published: May 6, 2021

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